

ENTHUSIAST ADVANCE COURSE

PHASE : MEC-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 18-06-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	2	2	2	3	2	1	4	4	3	3	3	1	4	3	3	2	2	2	4	3	4	3	1	2	2	3	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	1	4	3	2	1	2	3	4	3	3	3	1	4	2	2	3	3	3	2	1	1	1	1	3	3	2	4	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	3	2	2	2	3	2	4	3	2	1	3	1	4	3	3	2	3	2	2	1	2	3	1	1	4	4	3	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	3	2	1	2	2	1	4	4	1	4	3	4	2	2	4	3	2	1	4	4	4	1	1	3	4	3	1	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	4	1	1	1	3	3	2	2	1	2	2	1	3	3	1	2	2	4	1	3	2	3	2	2	2	2	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	1	2	3	2	4	1	1	2	1	2	3	2	4	4	2	3	4	3	1	1	2	2	2	1	2	4	4	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	1	2	2	4	4	1	2	2	2	3	4	2	4	2	2	3	1	2	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (1)

$$\frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2} = -\frac{0.02}{0.03} = -\frac{2}{3} \quad \dots (1)$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{-30} \quad \dots (2)$$

By solving equation (1) & (2)

$$f_1 = -10 \text{ cm}, f_2 = +15 \text{ cm}$$

2. Ans (4)

The separation between the objective and the eye

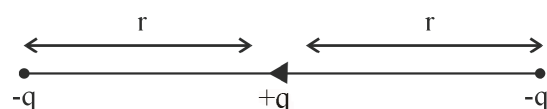
piece = Length of the telescope tube $f = f_0 + f_e$

$$\text{Here, } f_0 = 144 \text{ cm} = 1.44 \text{ m}$$

$$f_e = 6.0 \text{ cm} = 0.06 \text{ m}$$

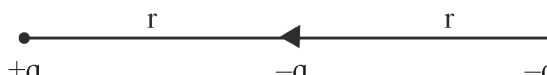
$$f = 1.44 + 0.06 = 1.5 \text{ m}$$

13. Ans (3)



$$U_1 = \frac{k(-q) \times q}{r} \times 2 + \frac{k(-q) \times (-q)}{2r}$$

$$U_1 = \frac{-3kq^2}{2r}$$



$$U_2 = \frac{k(-q) \times (-q)}{r} + \frac{k(-q) \times q}{2r} + \frac{k(-q) \times (-q)}{r}$$

$$U_2 = -\frac{kq^2}{2r}$$

$$U_1 : U_2 = 3$$

14. Ans (1)

$$\vec{E} = \left[\frac{\partial}{\partial x} v_i \hat{i} + \frac{\partial}{\partial y} v_j \hat{j} + \frac{\partial}{\partial z} v_k \hat{k} \right]$$

$$\vec{E} = - \left[5\hat{i} + 4\hat{j} - 6\hat{k} \right]$$

$$\vec{E} = -5\hat{i} - 4\hat{j} + 6\hat{k}$$

15. Ans (4)

They will not experience any force if

$$F_G = F_e$$

$$\Rightarrow G \frac{m^2}{(16 \times 10^{-2})^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{(16 \times 10^{-2})^2}$$

$$\Rightarrow \frac{q}{m} = \sqrt{4\pi\epsilon_0 G}$$

16. Ans (3)

Let final charge on the inner sphere be 'x' then

$$0 = V_{in} = \frac{kx}{R} + \frac{k(3Q)}{2R}$$

$$x = \frac{-3}{2}Q$$

17. Ans (3)

EPE = Charge $\times$ Potential

$$= (-q) \left(\frac{kQ}{3R} \right) = \frac{-kQq}{3R}$$

18. Ans (2)

If potential of both sphere are same then there will be No flow of charge

$$\text{or } V_1 = V_2$$

$$\text{or } K \frac{2q}{a} = K \frac{3q}{b}$$

$$\text{or } b = \frac{3}{2}a$$

19. Ans (2)

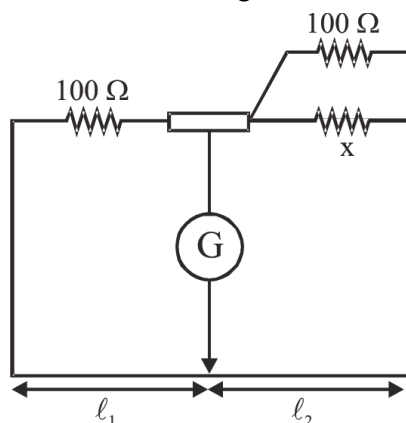
$$V_0 = \frac{KQ_{net}}{R}$$

$$Q_{net} = 3Q + Q - 2Q = 2Q$$

$$V_0 = \frac{2KQ}{R}$$

20. Ans (2)

$\therefore$ Wheatstone bridge is in balanced condition

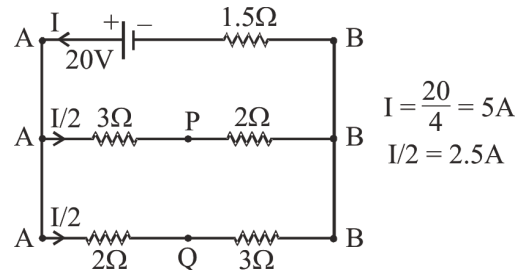


$$\text{So, } \frac{100}{l_1} = \frac{100x}{100+x}$$

$$\therefore \frac{l_1}{l_2} = 2$$

$$\Rightarrow x = 100 \Omega$$

21. Ans (4)



$$I = \frac{20}{4} = 5A$$

$$I/2 = 2.5A$$

$$V_A - V_P = (2.5)3 = 7.5V$$

$$V_A - V_Q = (2.5)2 = 5V \quad (\text{On subtraction})$$

$$\Rightarrow V_Q - V_P = 2.5V \Rightarrow V_Q > V_P$$

22. Ans (3)

The current through both the branches containing resistor is $\frac{10}{4} = 2.5A$

$$V_C - V_A = 1 \times 2.5 = 2.5V \text{ and}$$

$$V_C - V_B = 3 \times 2.5 = 7.5V \text{ Volt}$$

$$\text{So, } V_A - V_B = 5V \text{ Volt}$$

23. Ans (4)

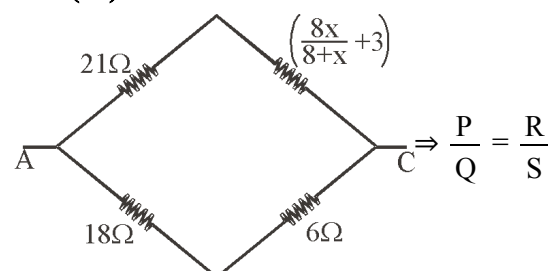
$$P = \frac{V^2}{R} \text{ so, } R = \frac{V^2}{P} \therefore R_1 = \frac{V^2}{100} \text{ \& } R_2 = R_3 = \frac{V^2}{60}$$

$$\text{Now, } P_1 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_1 \text{ and}$$

$$P_2 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_2 \text{ and } P_3 = \frac{(250)^2}{R_3}$$

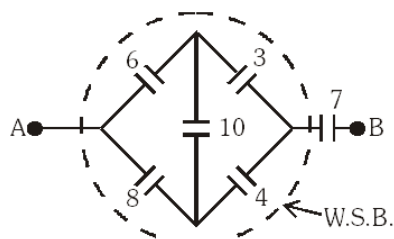
$$P_1 : P_2 : P_3 = 15 : 25 : 64 \Rightarrow P_1 < P_2 < P_3$$

24. Ans (3)



25. Ans (1)

Redraw circuit



C_{net} of W.S.B.

$$= \frac{6 \times 3}{6 + 3} + \frac{8 \times 4}{8 + 4}$$

$$= 2 + \frac{32}{12} = \frac{14}{3}$$

This is in series with 7

$$\text{So, } C_{\text{eq.}} = \frac{\frac{14}{3} \times 7}{\frac{14}{3} + 7} = \frac{14}{5}$$

26. Ans (2)

$$F = QE = Q \left(\frac{\sigma}{2\epsilon_0} \right) = \frac{Q^2}{2A\epsilon_0}$$

27. Ans (2)

The situation is shown in figure, when the switch is closed the charge flows from $2 \mu\text{F}$ capacitor to $4 \mu\text{F}$ capacitor till both acquires the same potential.

Let final common potential of capacitors is V_C , then from conservation of charge.

$$2 \mu\text{F} \times 4\text{V} = 2 \mu\text{F} \times V_C + 4 \mu\text{F} \times V_C$$

$$\Rightarrow V_C = \frac{8}{6} \text{V} = \frac{4}{3} \text{V}$$

So total final energy stored in capacitors is,

$$\sum U_f = \frac{2 \times V_C^2}{2} + \frac{4 \times V_C^2}{2} \mu\text{J}$$

$$\sum U_f = \frac{6}{2} \times \left(\frac{4}{3} \right)^2 \mu\text{J} = \frac{16}{3} \mu\text{J}$$

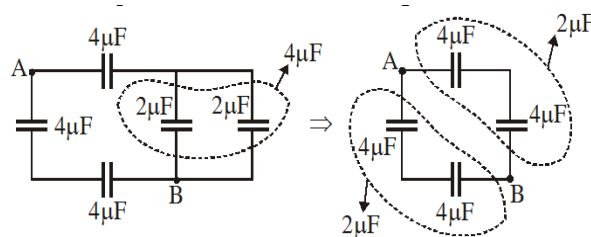
Note : The energy dissipated is equal to,

$$\sum U_i - \sum U_f = 16 - \frac{16}{3} = \frac{32}{3} \mu\text{J}$$

28. Ans (3)

TG: @Chalnaayaaar

The given circuit can be simplified as follows



Equivalent capacitance between A and B is

$$C_{AB} = 4 \mu\text{F}$$

30. Ans (4)

Interchanging cell and galvanometer do not effect balance condition.

33. Ans (1)

I case

$$\frac{10}{R} = \frac{\ell}{100 - \ell} \quad \dots (i)$$

$$\text{II case } \frac{10}{2R} = \frac{3/4 \ell}{100 - 3/4 \ell} \quad \dots (ii)$$

From (i) and (ii)

$$2 = \frac{4}{3} \frac{100 - 3/4 \ell}{100 - \ell} \Rightarrow \ell = \frac{200}{3}$$

Put the value of ℓ in equation (i)

$$\frac{10}{R} = \frac{200/3}{100 - \frac{200}{3}} = 2 \Rightarrow R = 5 \Omega$$

34. Ans (4)

Refer Module-04, Pg. # 126, Q.01

35. Ans (3)

$$\text{Reading of voltmeter} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$= 14 \text{ volt}$$

SECTION-B

36. Ans (2)

$$F_0 = 4 \text{ cm}, F_e = 6 \text{ cm}, u_0 = 6 \text{ cm}$$

$$\text{M.P.} = m_0 \left(1 + \frac{D}{F_e} \right)$$

$$\text{M.P.} = \left(\frac{f_0}{f_0 + u_0} \right) \left(1 + \frac{D}{f_e} \right)$$

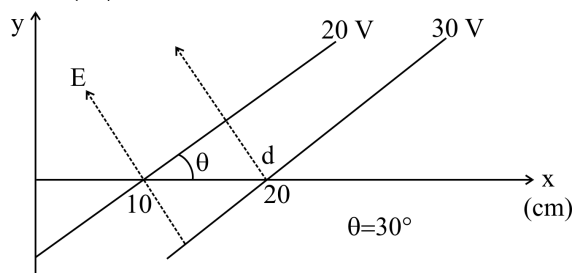
$$\text{M.P.} = \left(\frac{4}{4 - 6} \right) \left(1 + \frac{25}{6} \right)$$

$$\text{M.P.} = (-2) \times \frac{31}{6} = -10.33$$

38. Ans (2)

$$\therefore \sin \theta = \frac{n\lambda}{a}$$

41. Ans (3)



$$d = (20 - 10) \sin 30^\circ = 5 \text{ cm}$$

$$|\vec{E}| = \left| \frac{d\vec{V}}{dr} \right| = \frac{30 - 20}{5 \times 10^{-2}} = 200 \text{ Vm}^{-1}$$

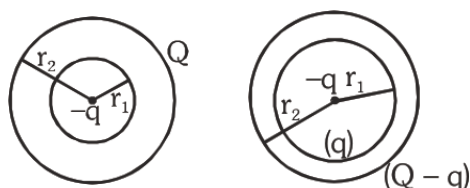
direction with x-axis

$$\alpha = \theta + 90 = 30 + 90$$

$$\alpha = 120^\circ$$

42. Ans (3)

Due to induction charge induced on shell



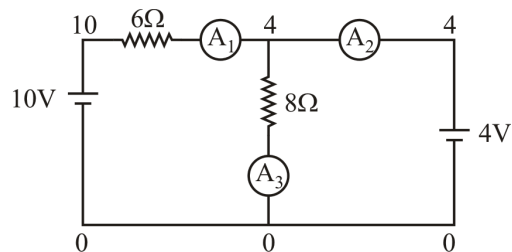
Before Induction

After Induction

$$\text{inner } \sigma = \frac{q}{4\pi r_1^2}$$

$$\text{outer } \sigma = \frac{Q - q}{4\pi r_2^2}$$

43. Ans (3)



$$A_3 \Rightarrow \frac{4 - 0}{8} = 0.5 \text{ A}$$

$$A_1 \Rightarrow \frac{10 - 4}{6} = 1 \text{ A}$$

$$A_2 \Rightarrow 1 - 0.5 = 0.5 \text{ A}$$

44. Ans (1)

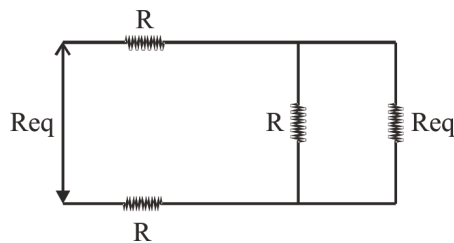
$$\text{By Solving } R_{AB} = \frac{8R}{11}$$

45. Ans (4)

TG: @Chalnaayaaar

To obtain balance point on potentiometer wire positive terminals of cells in secondary circuit must be connected with positive end of potentiometer wire.

46. Ans (2)



$$R_{eq} = r, \frac{rR}{r + R} + 2R = r$$

$$rR + 2Rr + 2R^2 = r^2 + Rr$$

$$r^2 - 2Rr - 2R^2 = 0$$

$$r = \frac{-(-2R) \pm \sqrt{4R^2 + 8R^2}}{2} = \frac{2R \pm \sqrt{12}R}{2}$$

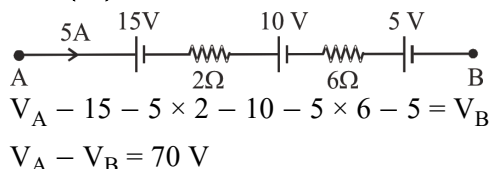
$$r = (1 + \sqrt{3}) R = 2(1 + \sqrt{3}) k\Omega$$

47. Ans (2)

Since current $I = neAv_d$ through both rods is same

$$2(n)eAv_L = ne(2A) v_R \text{ or } \frac{V_L}{V_R} = 1$$

48. Ans (3)



49. Ans (3)

$$R'_1 = R_1(1 + \alpha_1 \theta); R'_2 = R_2(1 + \alpha_2 \theta)$$

$$R_{net} = R'_1 + R'_2 = R_1(1 + \alpha_1 \theta) + R_2(1 + \alpha_2 \theta)$$

$$= R_1 + R_2 + \theta(R_1 \alpha_1 + R_2 \alpha_2)$$

$$= (R_1 + R_2) \left[1 + \frac{R_1 \alpha_1 + R_2 \alpha_2}{R_1 + R_2} \theta \right]$$

$$\text{Compare this equation } R = R_0(1 + \alpha_{net} \theta)$$

$$\therefore \alpha_{net} = \frac{R_1 \alpha_1 + R_2 \alpha_2}{R_1 + R_2}$$

50. Ans (3)

In case of charging of battery

$$TPD = E + Ir$$

$$= 18 + (9)(4) = 54 \text{ Volt}$$

SUBJECT : CHEMISTRY

SECTION-A

51. **Ans (2)**

NCERT XIIth Pg.#46 Para-I

56. **Ans (3)**

$$\frac{P^0 - P_s}{P_s} = \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$$

$$\frac{0.850 - 0.845}{0.845} = \frac{0.5}{\left(\frac{39}{78}\right) M_B}$$

$$\frac{0.005}{0.845} = \frac{0.5 \times 2}{M_B}$$

$$\boxed{M_B = 169}$$

63. **Ans (3)**

$$\therefore P_{\text{ideal}} = P_A^0 X_A + P_B^0 X_B$$

and solution of $\text{HNO}_3 + \text{H}_2\text{O}$ shows -ve deviation from Raoult's law

$$\therefore P_{\text{observed}} < P_{\text{ideal}}$$

64. **Ans (2)**

$$\therefore \pi_{\text{CH}_3\text{COOH}} = \pi_{\text{urea}}$$

$$\therefore (c \times i)_{\text{CH}_3\text{COOH}} = (c \times i)_{\text{urea}}$$

$$\frac{P}{60} \times 2 = \frac{Q}{60} \times 1$$

$$\frac{P}{Q} = \frac{1}{2}$$

65. **Ans (2)**

According to $(n + \ell)$ rule

$$4d - [4 + 2] = 6$$

$$4f - [4 + 3] = 7$$

$$4p - [4 + 1] = 5$$

66. **Ans (2)**

$$Z = 143 = \frac{\text{Unquadtrium}}{[\text{Uqt}]}$$

67. **Ans (3)**

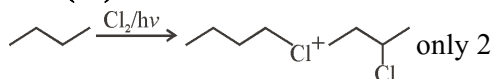
15^{th} group so Pnicogen

68. **Ans (2)**

Li show diagonal relationship in properties with magnesium.

74. **Ans (1)**

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SECTION-B

88. **Ans (4)**

$$\Delta T_f = i_m K_f = 1 \times \frac{1.8 \times 1.86}{180 \times 2} = 0.0093$$

91. **Ans (4)**

$$\therefore \Delta T_f = K_f \cdot m \cdot i$$

$$1.5 = 5 \times \left(\frac{12.2/122}{200} \times 1000 \right) \times i$$

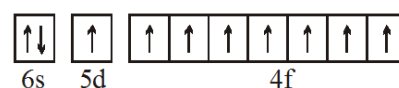
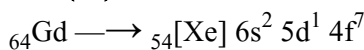
$$i = \frac{3}{5} = 1 + \alpha \left(\frac{1}{2} - 1 \right)$$

$$-\frac{2}{5} = -\frac{\alpha}{2}$$

$$\alpha = \frac{4}{5}$$

$$\% \alpha = \frac{4}{5} \times 100 = 80\%$$

93. **Ans (3)**



Total unpaired electrons = 1 + 7 = 8

94. **Ans (2)**

Fact

SUBJECT : BIOLOGY-I

SECTION-A

102. **Ans (4)**

NCERT XII Page No. 133

109. **Ans (2)**

NCERT XII Pg # 139/150(H) Fig : 7.10

110. **Ans (1)**

Both are mammals. They have similar structure of forelimbs and are a result of divergent evolution.

114. **Ans (1)**

NCERT (XIIth) Pg. # 129

115. **Ans (1)**

NCERT (XIIth) Pg. # 140

SECTION-B

141. **Ans (1)**
NCERT XII, Pg. No. 127
142. **Ans (3)**
NCERT XII, Pg. No. 130
143. **Ans (2)**
NCERT XII, Pg. No. 140
144. **Ans (3)**
NCERT XII, Pg. No. 141
145. **Ans (2)**
NCERT XII, Pg. No. 136

SUBJECT : BIOLOGY-II**SECTION-A**

160. **Ans (2)**
NCERT Pg. # 72-73 [E]
NCERT Pg. # 72-73 [H]
161. **Ans (1)**
During the formation of F_2 - generation both the characters appear, so no blending.

168. **Ans (3)**
NCERT Pg. # 79
180. **Ans (2)**
NCERT-XII, Pg. # 167-168
181. **Ans (1)**
NCERT Pg#168
182. **Ans (1)**
NCERT Pg. # 167
183. **Ans (2)**
NCERT PAGE : 166, 167, 169

SECTION-B

188. **Ans (2)**
NCERT Pg. 80 [E]
NCERT Pg. 80 [H]
189. **Ans (2)**
NCERT Pg#77
190. **Ans (2)**
NCERT XII Pg. # 89
198. **Ans (1)**
NCERT-XII, Pg#182, Para-9.1.2

TG: @Chalnaayaaar